Agentic AI Can Generalize to Multiple Speeded Tasks with Human-Like Behavior

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Introduction: There is growing evidence that online experiments may be contaminated by AI-based bots. One source of data that is particularly susceptible to AI bots are self-report surveys, which involve verbal prompts and verbal responses with minimal emphasis on response speed, making them particularly susceptible to large language model (LLM)-based bots. Westwood (2025) recently showed that an LLM-based synthetic respondent could pass attention checks, produce a coherent demographic persona with memory of its past responses, and produce plausible survey based responses on a variety of prompts. This suggested that self-report survey results from online experiments may already be contaminated by bots.

Recent work has suggested that even speeded tasks may be susceptible to agentic AI bots, but this existing work has key shortcomings. Huskey et al. (2026) recently used a Claude Opus 4.6 (Anthropic) based bot to complete the Attention Network Task (Fan et al., 2002), a prominent speeded RT task, with performance that matched many patterns in human data. However, this solution was tailored to a single online task, required considerable training and iteration, and some key bot behaviors diverged from humans (e.g., long tails of the RT distribution). In a brief preprint and a set of demonstration videos, Ozudogru et al. (2026) showed that a prompt-only approach could produce human-like RT distributions and mean RTs in 3 different cognitive tasks. However, they provide minimal explanation of the bots architecture and minimal evaluation of the correspondence with human behavior.

Here, we present an agentic AI bot that addresses these shortcomings with minimal training and iteration, has the ability to generalize across multiple tasks and multiple implementations of each task, and it reproduces specific human behavioral patterns including trial-by-trial sequential effects.

Methods: Given only an experimental URL, our agentic AI bot scrapes the page source and runtime JavaScript state and passes this to Claude Opus 4.6 (Anthropic) in a single API call. Claude analyses the source by identifying the task paradigm and generating a task schema that specifies: response distributions (ex-Gaussian mu, sigma, and tau per experimental condition), per-condition accuracy and omission rate targets, sequential RT effects, and between-subject performance variance. Parameter selection is determined by Claude's knowledge of the task literature and without any task-specific prompting from the researcher.

Two aspects constitute the researcher's contribution on bot performance: the choice of distribution family (ex-Gaussian) and selection of six available sequential RT effects (autocorrelation, fatigue drift, post-error slowing, condition repetition, 1/f noise, and post-stop-signal slowing) from which Claude selects and parameterizes as appropriate for the identified task. A brief automated pilot session ($\sim$20 trials) validates Claude's stimulus detection selectors against the live experiment DOM, triggering a refinement loop if selectors fail. The resulting configuration is cached and reused for all subsequent runs without additional API calls.

We evaluated bot performance on four implementations of two tasks: stop signal (Experiment Factory, N=29 bot "subjects"; STOP-IT, N=27 bot "subjects") and Stroop (Experiment Factory, N=29 bot "subjects"; Cognition.run, N=28 bot "subjects"), comparing bot performance to human reference data from Eisenberg et al. (2019), which acquired stop signal (Logan & Cowan, 1984) and Stroop (1935) data in 522 online subjects via MTurk.

Results: For the stop signal task, bot performance on the Experiment Factory (Sochat et al., 2016) implementation fell within the human distribution on go RT (bot: 622 ± 47 ms; human: 585 ± 73 ms; $z = +0.52$) and stop accuracy (bot: $53.2 \pm 2.2\%$; human: $52.4 \pm 2.2\%$; $z = +0.37$). Estimated SSRT was faster than the human mean (bot: 214 ± 100 ms; human: 287 ± 53 ms; $z = -1.37$), but SSRT in this sample is longer than typical SSRTs (e.g., simple stop SSRT ≈ 235ms in Bissett et al., 2023). This pattern replicated across the STOP-IT implementation without modification to the bot's configuration, demonstrating cross-platform generalization. For the Stroop task, congruent RT (bot: 706 ± 42 ms; human: 672 ± 102 ms; $z = +0.33$), incongruent RT (bot: 798 ± 56 ms; human: 795 ± 123 ms; $z = +0.02$), and the Stroop interference effect (bot: 93 ± 40 ms; human: 123 ± 61 ms; $z = -0.50$) all fell within the human distribution. In addition to mean-level metrics, the bot produced positive lag-1 RT autocorrelation (stop signal: $r = .06 \pm .09$; Stroop: $r = .09 \pm .17$) and post-error slowing (stop signal: 16 ± 28 ms; Stroop: 17 ± 165 ms), comparable in direction to the human data (autocorrelation — stop signal: $r = .21 \pm .14$; Stroop: $r = .09 \pm .12$; post-error slowing — stop signal: 1 ± 51 ms; Stroop: 59 ± 136 ms). In contrast to Ozudogru et al. (2026), these comparisons are made against a large human reference dataset (stop signal N = 447; Stroop N = 502; Eisenberg et al., 2019) using z-scores relative to the human distribution and not visual inspection of RT histograms.

Figure 1: Bot positioning within human distributions. Violin plots show the human reference distributions from Eisenberg et al. (2019); colored dots represent individual bot "subjects" across four task implementations (top row: stop signal; bottom row: Stroop).

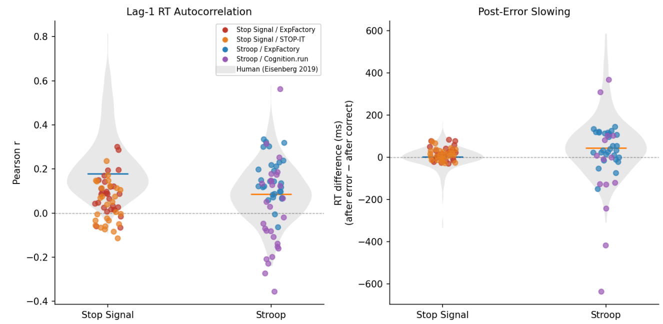

Figure 2: Sequential effects in bot and human data. Left panel: lag-1 RT autocorrelation; right panel: post-error slowing. Gray violins show the human reference distribution; colored dots represent individual bot "subjects" across task implementations.

Conclusion: This work shows that current agentic AI tools, with minimal human iteration, can produce human-like performance across multiple tasks. This proof of concept shows that real-world data acquisition in speeded cognitive tasks may be susceptible to bot contamination today.

Acknowledgments: The agentic bot described in this work uses Claude Opus 4.6 (Anthropic) for task identification and parameter generation.

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